Cyclohexane Reactor Ver 1 · operator training simulator

Hazardous-area classification schedule

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Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- 20 sources of release in 6 hazardous areas; zones and extents per typical practice for open-air plant (IEC 60079-10-1 / API RP 505 style, assumed).
- Gas group IIC wherever hydrogen is present; temperature class T3 wherever cyclohexane (AIT ≈ 245 °C) is present.
- From the model: the R-101 liquid at 200 °C flashes 80 % on release; the R-101 overhead gas is 43 % cyclohexane vapour and 1.32 × as dense as air.
- Hot surfaces of the model checked against the T-class limit and the auto-ignition temperatures (section 5).

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Basis

Item	Basis
Method	Sources of release, grade, ventilation → zone and extent, in the style of IEC 60079-10-1 and API RP 505; extents are typical practice for open-air plant (assumed), not a computed release.
Ventilation	Open-air structure: natural ventilation, adequate dilution, good availability (assumed); analyzer shelter mechanically ventilated.
Grades	Continuous: none in the unit (no open vents of flammables). Primary: openings used in normal operation (sampling, catalyst charging, filter change-out, drain pits). Secondary: seals, flanges, packing that leak only by failure.
Zones	Primary → Zone 1 near the opening with Zone 2 round it; secondary → Zone 2. Pits and trenches inside a Zone 2 become Zone 1 (heavier-than-air vapour).
Gas group	IIC where hydrogen is present (most demanding: smallest ignition energy and flame gap); IIA for benzene and cyclohexane alone.
Temperature class	The equipment surface limit must stay below the lowest auto-ignition temperature (AIT) in the area: benzene ≈ 498 °C, cyclohexane ≈ 245 °C, hydrogen ≈ 560 °C (typical literature values) → T3 (200 °C) wherever cyclohexane is present.
Equipment protection level	Zone 0 Ga · Zone 1 Gb · Zone 2 Gc (IEC 60079-14).

2 · Substances (typical literature values)

Substance	Flash point °C	Normal boiling point °C	Auto-ignition °C	LFL–UFL % vol	Vapour density (air = 1)	Gas group	T-class needed
Benzene	-11	80.1	498	1.2–7.8	2.7	IIA	T1
Cyclohexane	-20	80.7	245	1.3–8.4	2.9	IIA	T3
Hydrogen	gas	-253	560	4–75	0.07	IIC	T1
Methane (make-up impurity)	gas	-162	537	5–15	0.55	IIA	T1

3 · Fluids of the unit

Key	Fluid	Where	Release category	Group	Lowest AIT °C	T-class	From the model (design)
B	Benzene feed liquid	TK-100 → P-100A/B → FV-101 / XV-101 → R-101	C: liquid above its flash point, below its boiling point (pool and vapour)	IIA	498	T1	40 °C: below the 80.1 °C boiling point, does not flash; evaporates from a pool
C	R-101 slurry and pump-around (cyclohexane, ≈2 % benzene, Raney-Ni)	R-101, P-101A/B, E-101 tubes, FV-104, LV-101	A: flashing liquid (above its boiling point at 1 atm)	IIA	245	T3	200 °C, 30.0 bar g: 80 % flashes on release ($c_p \cdot (T - T_b)/\lambda$ with the model's cpL and λ at 80.7 °C)
C2	R-102 slurry and transfer liquid	R-102, P-102A/B, F-102A/B, LV-102 → R-103	A: flashing liquid	IIA	245	T3	160 °C, 29.6 bar g: 53 % flashes
P	Cyclohexane product	R-103 outlet, E-106, V-104, LV-104 → TK-106 / TK-107, sample point	C: liquid above its flash point	IIA	245	T3	45 °C: below its boiling point, above its flash point
Hc	R-101 overhead gas (hydrogen + cyclohexane vapour)	R-101 vapour space → E-103 / V-103	G: gas, here heavier than air	IIC	245	T3	43 % cyclohexane vapour, MW 38.1 → 1.32 × air: heavier than air
Hv	E-103 vent gas to R-102	V-103 → R-102 sparger → R-103	G: gas	IIC	245	T3	after E-103 at 55.7 °C: 1.5 % cyclohexane, MW 4.3 (0.15 × air)
H	Recycle and make-up hydrogen	V-104 → K-101 → FIC-103; PV-105 make-up; HIC-107 purge	G: buoyant gas	IIC	245	T3	at V-104 45 °C: 1.1 % cyclohexane, H_2 92.1 %, MW 4.0 (0.14 × air): buoyant

4 · Schedule of sources of release

Extents are radii from the source (m); 'grade layer' is a 0.6 m high zone along grade to the stated distance (heavier-than-air vapour). Ventilation: open air, adequate, good availability (assumed).

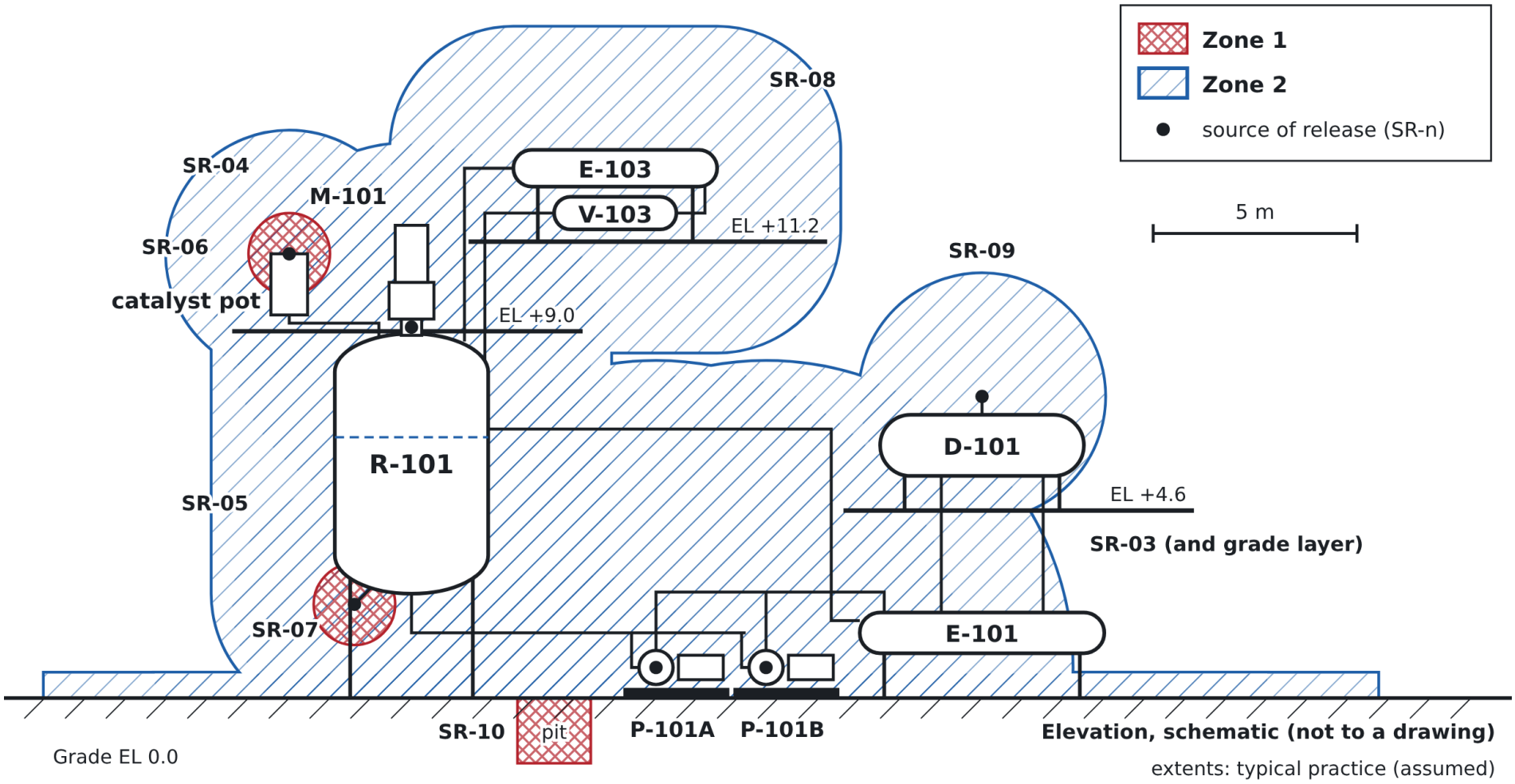
SR	Area	Equipment / line	Source of release	Fluid	Grade	Zone	Z1 R m	Z2 R m	Grade layer m	Note
SR-01	HA-1	P-100A/B benzene feed pumps	mechanical seals (dual, assumed)	B	Secondary	2	–	4.5	7.5 × 0.6	heavier-than-air vapour spreads along grade
SR-02	HA-1	FV-101 / XV-101 / FT-101 feed station	flanges, valve stem packing	B	Secondary	2	–	3	–	
SR-03	HA-2	P-101A/B pump-around pumps	mechanical seals (dual, assumed)	C	Secondary	2	–	7.5	15 × 0.6	flashing liquid at 200 °C: larger extent
SR-04	HA-2	M-101 agitator on R-101	top-entry shaft seal (dual pressurised, assumed)	Hc	Secondary	2	–	4.5	–	R-101 vapour-space gas is heavier than air
SR-05	HA-2	R-101 nozzles, manway, instrument flanges	flanged joints	C / Hc	Secondary	2	–	3	–	envelope round the vessel
SR-06	HA-2	Catalyst addition pot (HV-117)	charging connection opened in normal operation	C	Primary	1 / 2	1	3	–	Raney-Ni under cyclohexane and nitrogen
SR-07	HA-2	R-101 bottom drain and sample connection	sampling, draining	C	Primary	1 / 2	1	3	–	closed sampler recommended
SR-08	HA-2	E-103 / V-103 reflux condenser and drum	flanged joints	Hc	Secondary	2	–	3	–	
SR-09	HA-2	D-101 steam-drum vents and blowdown	hydrocarbon only after an E-101 tube leak	C	Secondary	2	–	3	–	tube side 30 bar g into a 4.5 bar g drum
SR-10	HA-2	Closed-drain pit and trenches below grade	drains, heavier-than-air vapour collecting	B / C	Primary	1 / 2	–	3	–	Zone 1 inside the pit
SR-11	HA-3	P-102A/B transfer pumps	mechanical seals (dual, assumed)	C2	Secondary	2	–	7.5	15 × 0.6	flashing liquid at 160 °C
SR-12	HA-3	F-102A/B catalyst filters	drain and vent at each change-out	C2	Primary	1 / 2	1.5	4.5	–	drained and nitrogen-purged before opening
SR-13	HA-3	R-102 nozzles and instrument flanges	flanged joints	C2 / Hv	Secondary	2	–	3	–	jacket side (steam / CW) is not a source
SR-14	HA-4	R-103, E-106, V-104 flanges	flanged joints	P / H	Secondary	2	–	3	–	
SR-15	HA-4	V-104 level bridle and LV-104 station	drains, stem packing	P	Secondary	2	–	3	–	
SR-16	HA-4	Product sample point (laboratory check of AI-103 / AI-104)	sampling	P	Primary	1 / 2	1	3	–	
SR-17	HA-5	K-101 recycle compressor	shaft seals / packing; distance-piece vents to the flare (assumed)	H	Secondary	2	–	7.5	–	buoyant gas: extent mainly upward; open-sided shelter
SR-18	HA-5	PV-105 make-up and HIC-107 purge stations	flanges, stem packing	H	Secondary	2	–	3	–	
SR-19	HA-6	Analyzer shelter (AI-100, AI-103, AI-104, AI-105)	sample-conditioning fittings inside	H / P	Secondary	2	–	1	–	Zone 2 inside with ventilation ≥ 12 air changes/h and gas detection (assumed); 1 m round openings
SR-20	—	Relief valves, EDV-101, HV-121 vent	discharge to the closed flare header	H / C	—	none in the unit	–	–	–	flare FL-100 outside the battery limit

5 · Hazardous areas, gas group and temperature class

Required T-class: the hottest class whose surface limit is below the lowest AIT in the area (T1 450, T2 300, T3 200, T4 135 °C). Hot process surfaces are compared with the limit at design and with the AIT at the upper bound the safety system allows. An AIT is measured in a closed heated flask; an open hot surface usually needs a higher temperature, so the comparison is conservative.

Area	Fluids	Group	Lowest AIT °C	T-class (limit °C)	Limit < AIT	Hot surfaces: design → upper bound °C	Max at design °C	Design ≤ limit	Upper bound °C	Upper bound < AIT
HA-1 Benzene feed pumps and feed station	B	IIA	498	T1 (450)	✓	benzene feed 40	40.0	✓	40.0	✓
HA-2 R-101 structure: R-101, M-101, P-101A/B, E-101, D-101, E-103, V-103	C, Hc, B	IIC	245	T3 (200)	✓	R-101 and pump-around 200 → 220 (TSHH-100 setting) R-101 if the trip fails 200 → 258 (boiling ceiling at PI-101) MP steam on E-101 (HEAT) 184 D-101 156	200.0	✓	258.3	✗ EQ-ee-4
HA-3 R-102, P-102A/B, F-102A/B	C2, Hv	IIC	245	T3 (200)	✓	R-102 160 → 220 (R-101 trip setting (feed from R-101)) R-102 jacket MP steam 184	184.0	✓	220.0	✓
HA-4 R-103, E-106, V-104	C2, P, H	IIC	245	T3 (200)	✓	R-103 hottest bed 159 → 220 (TSHH-107 setting) V-104 45	158.6	✓	220.0	✓
HA-5 K-101, make-up and purge stations	H	IIC	245	T3 (200)	✓	K-101 discharge (estimate, η 0.75) 57 K-101 suction 45	56.6	✓	56.6	✓
HA-6 Analyzer shelter	P, H	IIC	245	T3 (200)	✓	sample lines 45	44.9	✓	44.9	✓

6 · Zone sketch of the R-101 area (elevation, schematic)



Secondary sources (seals, flanges) give Zone 2; openings used in normal operation (catalyst pot, drain and sample point) a Zone 1 inside a Zone 2; the closed-drain pit is Zone 1. The flashing pump-around liquid gives the widest zone and a layer along grade. Extents from section 4; layout schematic, not to a drawing.

7 · Equipment protection recommendations

Equipment	Where	Recommended protection	Why
LV motors (P-100, P-101, P-102 and auxiliaries)	Zone 2 (HA-1 … HA-5)	Ex ec IIC T3 Gc (Ex eb or Ex db acceptable)	Zone 2 needs EPL Gc; IIC because hydrogen is present in HA-2 … HA-5; T3 because cyclohexane (AIT ≈ 245 °C) is present. One plant-wide rating IIC T3 keeps one spare stock; HA-1 alone would allow IIA T1.
MV motors (K-101 and, at a 200 kW threshold, M-101)	Zone 2 (HA-2, HA-5)	Ex pzc (pressurised, pre-start purge) or Ex ec with a rotor/stator sparking assessment	Large MV motors can spark in the air gap at start; IEC 60079-7 asks for a risk assessment or a pre-start purge; anti-condensation heater also T3.
Motors in Zone 1	none: keep motors outside the 1–1.5 m Zone 1 envelopes (catalyst pot, drains, sample points, filter vents)	if unavoidable: Ex db or Ex eb IIC T3 Gb	Zone 1 needs EPL Gb.
Transmitters, switches, positioners, thermocouple heads	Zone 1 and 2	Ex ia IIC T4 Ga/Gb (intrinsic safety)	Live work without a gas-free permit; IIC barriers and cable parameters checked for hydrogen.
Solenoid valves (XV-101, XV-105, EDV-101, SOVs)	Zone 2	Ex mb or Ex db IIC T4, or low-power Ex ia	De-energise to trip; coil surface stays below the T3 limit.
Junction and terminal boxes	Zone 1 and 2	Ex eb IIC	Increased safety has no flame path, so IIC is easy (Ex d IIC enclosures need tighter flame paths and barrier glands).
Lighting fittings	Zone 1 and 2	Ex db eb IIC T3 (LED) in Zone 1; Ex ec IIC T3 in Zone 2	Surface temperature of the fitting below 200 °C.
Electric heat tracing	analyzer sample lines (HA-4, HA-6)	self-regulating, Ex e, T3 by stabilised design or a limiter	Sheath temperature must stay below 200 °C.
Gas detectors and flame detectors	HA-1 (benzene), HA-2 and HA-5 (hydrogen)	Ex db IIC T4 (catalytic or IR for hydrocarbons; catalytic or electrochemical for hydrogen)	IR does not see hydrogen; hydrogen detectors high near K-101 and the R-101 top.
Analyzer shelter	HA-6	Zone 2 rated equipment inside, or Ex p shelter with a purge-loss alarm	Sample fittings release inside a room; ventilation is what keeps the zone small.
Local control stations, push-buttons, sockets	Zone 2	Ex db eb IIC T3; sockets interlocked	Welding and portable equipment need a gas-free permit.
Cable entries	all	Ex e glands; barrier glands into Ex d IIC enclosures	IIC flame paths; inspection per IEC 60079-17.

8 · Checks

Check	Values	
T-class limit below the lowest AIT in every area	HA-1 T1, HA-2 T3, HA-3 T3, HA-4 T3, HA-5 T3, HA-6 T3	✓
HA-1: T-class limit ≥ hottest process surface at design	450 ≥ 40.0 °C	✓
HA-2: T-class limit ≥ hottest process surface at design	200 ≥ 200.0 °C	✓
HA-3: T-class limit ≥ hottest process surface at design	200 ≥ 184.0 °C	✓
HA-4: T-class limit ≥ hottest process surface at design	200 ≥ 158.6 °C	✓
HA-5: T-class limit ≥ hottest process surface at design	200 ≥ 56.6 °C	✓
HA-6: T-class limit ≥ hottest process surface at design	200 ≥ 44.9 °C	✓
HA-1: upper-bound surface below the AIT	40.0 vs 498 °C	✓
HA-2: upper-bound surface below the AIT	258.3 vs 245 °C	✗
HA-3: upper-bound surface below the AIT	220.0 vs 245 °C	✓
HA-4: upper-bound surface below the AIT	220.0 vs 245 °C	✓

Check	Values	
HA-5: upper-bound surface below the AIT	56.6 vs 245 °C	✓
HA-6: upper-bound surface below the AIT	44.9 vs 245 °C	✓
Gas group IIC wherever hydrogen is present	HA-2, HA-3, HA-4, HA-5, HA-6	✓
Every zone on the sketch comes from a source in the schedule	8 zones	✓
Every motor-driven machine of the load list has a seal source	P-100, P-101, P-102, K-101, M-101	✓
Fluid C is a flashing release (> 40 % flashes)	80 %	✓

9 · Engineering queries

The query found here (and the related SIS supply question) is listed in full in CHX-V1-EE-LL-001 section 11.

ID	Query	Documents	Proposed resolution
EQ-ee-4	The boiling ceiling of R-101 (TbC at PI-101 ≈ 258.3 °C) is above the auto-ignition temperature of cyclohexane (≈ 245 °C, typical literature value). Only TSHH-100 at 220 °C keeps the reactor and pump-around surfaces below it.	HAC-001 · Safety layers (critical limits) · SRS (SIF-1)	Keep R-101 and pump-around insulation and cladding intact (no oil-soaked lagging); record hot-surface ignition as a consequence in the HAZOP / LOPA of the runaway scenario; electrical equipment stays T3.
EQ-ee-6	EDV-101 is assumed fail-open and the SIS de-energises to trip: a loss of SIS power would open EDV-101 and depressure the hydrogen loop to the flare, although only the ESD opens it in the model.	Safety layers (SL_ASSUMED) · SRS · C&E · LL-001 UPS list	Decide in the SRS whether loss of SIS power is an ESD (EDV-101 opens) or a unit trip (EDV-101 energise-to-open with line monitoring, or held shut on a separate supply).

10 · Assumptions

Extents of each zone (typical practice for open-air plant, assumed) · Ventilation adequate with good availability · Dual mechanical seals on all pumps and the agitator · Distance-piece vents of K-101 to the flare · AIT, flash point, flammable limits: typical literature values · Analyzer-shelter ventilation ≥ 12 air changes per hour · Layout of the elevation sketch (schematic) · K-101 discharge temperature from k 1.4 and η 0.75.